

Lab 4

Course goals

- To implement and use data structures and algorithms in application programs: graph algorithms.
 - Shortest path algorithms.
 - Minimum spanning tree algorithms.

Preparation

Please ensure you read the entire lab description thoroughly, before starting any exercises, including the preparation steps given below. This lab has three parts. For Part 3, no code is provided; you are required to develop the entire C++ program from scratch.

Completing the following tasks before the Lab4 HA session will help you make the most of your lab time.

- Download the [files for this exercise](#) from the course website. Like previous labs, you can then use CMake to create a solution with two projects for this lab, one for the exercise in part 1 and another for the exercise in part 2.
- For each project, it's possible to compile, link, and execute the program. There is a `main.cpp` file with the `main` function, for each of the projects.
- Implement the graph algorithm requested in [Part 1](#).
- Read [Part 2](#) and understand the code given for this part.
- Read [Part 3](#) and prepare a description of the data structures, preferably using diagrams, along with pseudocode for the Dijkstra's algorithm. This will allow your lab assistant to review your approach before you begin coding.

Please note that in part 1 and part 2, there is no vertex labeled zero. Therefore, slot zero in data structures such as the `dist` vector should remain unused.

If you have any specific question about the exercises, then send us an e-mail. Be short and concrete, otherwise you won't get a quick answer. You can write your e-mail in Swedish. Add the course code to the e-mail's subject, i.e. "TND004: ...".

Part 1: shortest path algorithms

In this first part, you are requested to implement the UWSSSP algorithm (shortest path in an unweighted graph). To this end, follow the steps below.

- Review [lecture 12](#) and [lecture 13](#).
- Read sections 9.1 and 9.3.1 of the course book.
- Study carefully the code given for this exercise: `edge.h`, `digraph.h`, and `digraph.cpp`.
- **Implement** functions `Digraph::uwssp` and `Digraph::printPath`.
 - Use container `std::queue` for the implementation of function `Digraph::uwssp`.

An example of the input graph for the exercises in this part is shown below.

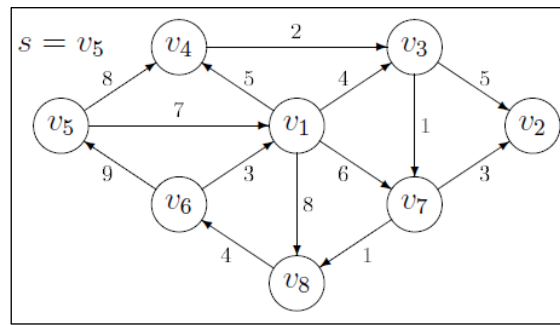


Figure 1: A (weighted) directed graph G and a start vertex s .

The corresponding output, a shortest path-tree, is exemplified in figure 2.

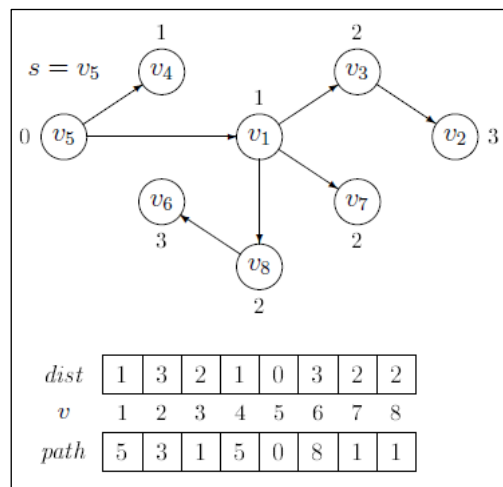


Figure 2: Unweighted shortest path-tree for start vertex s .

In this lab, graphs are represented with adjacency lists. This graph representation was introduced in [lecture 12](#). Note that the loop below (in pseudo-code) to iterate through the adjacency list of a vertex v (used in several of the graph algorithms you have seen during the lectures of the course)

```
for all  $(v,u) \in E$  do {
    ...;
}
```

can be implemented as (table is a `std::vector` with all vertices of the graph and `std::list` is used to store the adjacency lists)

```
for (auto& e : table[v]) {
    int u = e.to;    // e = (v,u)
    ...;
}
```

The files `digraph1_test_run.txt` and `digraph2_test_run.txt` contain an example of the program execution, for the graphs provided in the files `digraph1.txt` and `digraph2.txt`, respectively.

Part 2: minimum spanning tree algorithms

In this second part, you are requested to implement the Prim's and Kruskal's algorithms. To this end, follow the steps below.

- Review [lecture 14](#).

- Read sections [8.1-8.5] and 9.5 of the course book.
- Study carefully the code given for this exercise: `dsets.h`, `dsets.cpp`, `edge.h`, `graph.h`, and `graph.cpp`.
- Implement functions `Graph::mstPrim` and `Graph::mstKruskal` and test that they work as expected. Both functions build a minimum spanning tree¹ (MST) from a given connected weighted undirected graph $G = (V, E)$. They should display the edges of a minimum spanning tree and the total weight of the MST built. The implementation of `Graph::mstKruskal` requires a min-heap of vertices for which the `std::priority_queue` container can be useful.
- Re-implement function `Dsets::join` so that union by size is performed. Re-implement also function `Dsets::find` so that find with path compression is performed. The code distributed with this lab uses simple union and simple find for these functions.

Like as for [part 1](#), graphs are represented with adjacency lists.

Input and output for the Prim's and Kruskal's algorithms are exemplified in the figure below. In particular, the output (i.e. a MST) is essentially just a list of $|V| - 1$ graph edges.

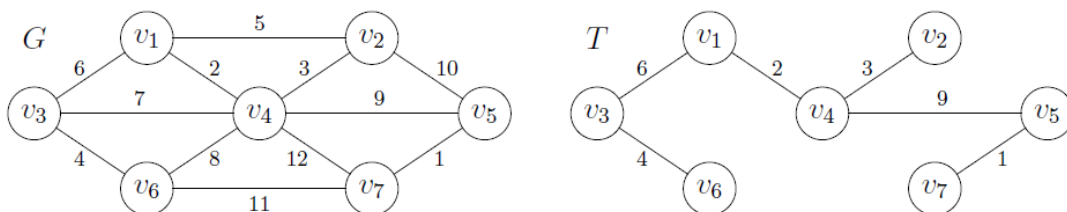


Figure 1: A connected weighted undirected graph G and a minimum spanning tree T for G (in this case, the tree is unique).

The files `graph1_test_run.txt` and `graph2_test_run.txt` contain an example of the program execution, for the graphs provided in the files `graph1.txt` and `graph2.txt`, respectively.

Part 3: travel planner

In this third part, you will implement a C++ program to plan flight itineraries that achieve the earliest possible arrival time at a destination airport, given a start airport and departure time, using Dijkstra's algorithm.

Your program must display a menu of options like the one below:

```
===== Menu =====
1. Read Graph
2. Print Graph
3. Dijkstra
4. Travel Plan
5. Print Earliest Arrival Paths
9. Quit
=====
Your choice ? |
```

- **Option 1:** Load flight data from a text file and construct the directed graph.

¹ i.e. a spanning tree for graph G of minimum total weight.

- **Option 2:** Display the graph by listing outbound flights for each airport.
- **Option 3:** Given a start airport s and time t , compute travel plans to all destinations (allowing stopovers), using only flights departing no earlier than t , and determining the earliest arrival time for each destination. Dijkstra's algorithm should be used.
- **Option 4:** Show the travel plan from s to a specified destination.
- **Option 5:** Show the travel plans from s to all reachable destinations.

A sample file, `flights-data.txt`, is provided. It begins with the number of airports, followed by a list of airport codes (one per line), and a common date for all flights. Each flight is listed on a separate line. The data is unsorted and may contain empty lines. All flights depart and arrive on the same day. Do not modify the file structure.

Use a heap-based priority queue (e.g., [`std::priority\_queue`](#)) to efficiently select the next airport to process, and represent the graph using adjacency lists.

A sample program run is provided in the file `travel-execution.txt`.

To obtain approval

For this exercise, students must demonstrate the following:

- Clearly describe the data structures used, preferably with diagrams showing their relationships, and be prepared to orally explain and justify your design choices. This includes describing
 - What the vertices and edges of the directed graph represent.
 - The meaning of the distance function in this context.
 - Whether the priority queue is implemented as a min-heap or max-heap, and the sorting criterion used.
 - The relaxation condition applied in your implementation of Dijkstra's algorithm.
- Present time and space complexity analysis of the implemented Dijkstra's algorithm (Option 3).

Presenting lab and deadlines

The exercises in this lab are compulsory and you should demonstrate your solutions during the lab session *Lab4 RE*. Read the instructions given in the [labs webpage](#) and consult the course schedule.

As usual, we expect that good programming practices are followed. Your code must be readable and well-indented. You are required to be able to explain the code you present. In addition, be prepared to discuss all points required for [approval of part 3](#).

If your solution for lab 4 has not been approved in the scheduled lab session *Lab4 RE* then it is considered a late lab. All groups have the possibility to present one late lab on the extra RE lab session scheduled in the end of the course.

Note that **we can only guarantee that each group can present one late lab**. In the extra RE lab session, priority is given to presentation of lab 4, then lab 3, and finally lab 2.